DS & AI Lab Project (BSDA4001)

TerraWatch: Intelligent Satellite Image Change Detection and Analysis

MILESTONE 3: MODEL SELECTION AND SYSTEM ARCHITECTURE

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BS in Data Science and Applications

G R O U P 7

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1. Introduction

TerraWatch identifies visible building and road changes between two satellite images and produces descriptions linked to visual evidence. Milestone 3 establishes the model choices, module interfaces, training configuration and complete input-to-output architecture [1]. The system combines pixel-level detection, domain-specific captioning, structured claims, crop-based evidence and an accept/reject/defer referee.

The data contract uses the Milestone 2 strict LEVIR-MCI package: **9,544 pairs**, partitioned into **6,444 training, 1,242 validation and 1,858 test records** after the documented exclusions. Source masks and five reference captions support supervised learning and evaluation. Inference consumes ordered images and model-generated intermediates. The local POC run below reads the original downloaded archive, which has a 1,929-pair test split; its subset results must not be presented as results on the cleaned 1,858-pair handover.

2. Overall System Architecture

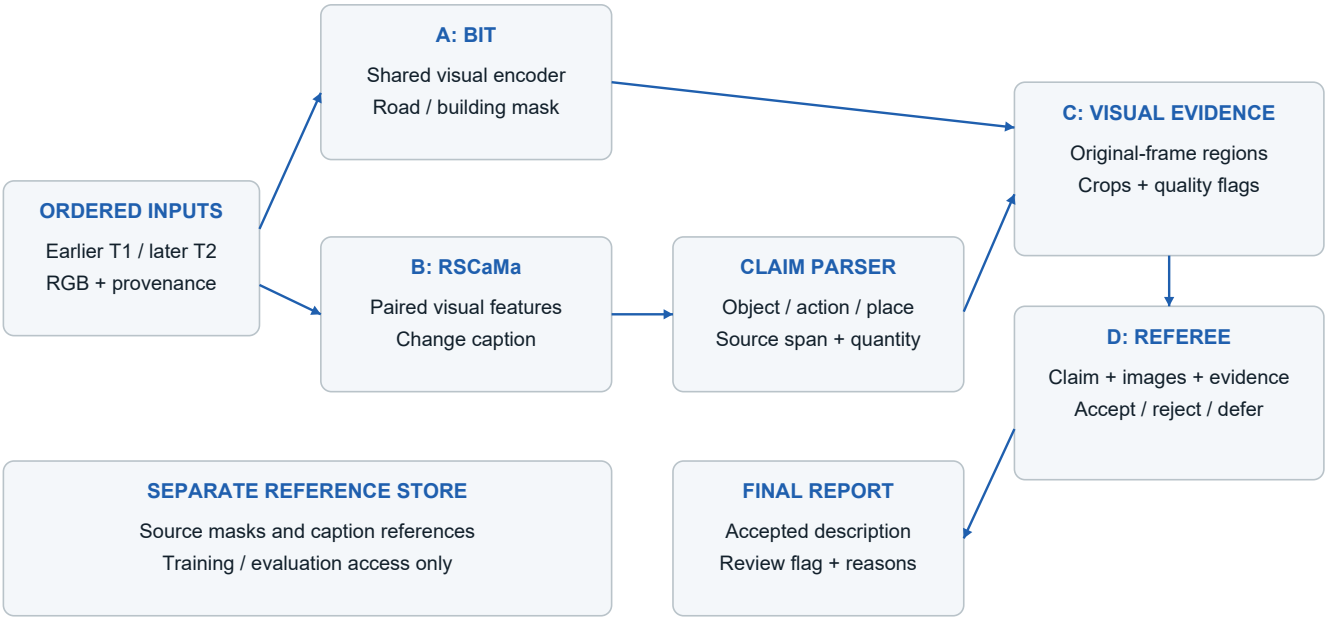


Figure 1. Detection and captioning operate on the same image pair. Predicted regions and caption claims meet at the evidence stage. Evaluation references occupy a separate store.

Layer	Technology and role
Data and persistence	Python, NumPy, Pillow; immutable image sources, split manifests, JSON/JSONL records and file hashes
Detection and description	PyTorch; BIT three-class detector and RSCaMa captioner in dedicated model environments
Evidence and decisions	OpenCV, Pydantic and isolated Transformers workers; explicit coordinates, typed records and decision gates
Execution and interaction	Colab/WSL/local CLI; two ordered image paths and a configuration file produce a JSON report and crop evidence

Model weights are acquired from their author repositories or Hugging Face. The inference design runs through local workers with cached weights. Evidence retrieval uses run IDs and file paths; the task does not require a vector database or document-retrieval stage.

3. End-to-End Workflow

Stage	Input	Output and control
Validate the pair	Earlier T1 and later T2 RGB images	Matching dimensions, temporal-order declaration and SHA-256 provenance; incompatible pairs raise an input error
Detect changes	Normalised temporal tensors	Background/road/building mask; nearest-neighbour restoration to original image dimensions
Generate and parse	Full-frame images; captioner output	Caption plus atomic object/action/location/quantity claims, source spans and parser provenance
Localise and inspect	Claim, predicted class mask and original images	Matching connected region, padded T1/T2 crops, transform record, visibility flags and structural observations
Referee and gate	Claim, fallible evidence and original images	Raw accept/reject/defer decision; probability validity and operating-threshold checks
Compose and store	Gated claim records	Accepted-description text, review flag, per-claim reasons, raw responses and auditable report JSON

Connected components identify candidate regions; they are not building-instance counts. Claim locations use the image-relative 3 x 3 grid. Matching regions retain original-T1 half-open pixel coordinates. Unlocalised claims use full-image context and receive a review reason.

The implementation executes each learned stage in a separate process, releases its memory on exit and records request, response and log files. A new run directory preserves earlier deliveries. Worker timeouts and missing referee outputs yield explicit error/review records. Insufficient visibility, an unsupported legal inference or an exact count without independent instance evidence produces a defer decision.

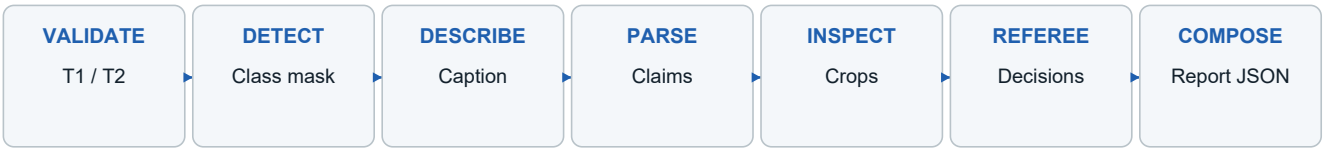


Figure 2. The operating pipeline retains intermediate evidence and raw decisions. Report composition uses accepted claim spans and carries review requirements into the final output.

4. Model Architecture Selection

Module	Selected architecture	Size, representation and integration
A - change detection	BIT with a shared ResNet18 backbone and three-class output head [2]	Four semantic tokens per temporal stream; one token-encoder layer and eight decoder layers. Class indices: 0 background, 1 road, 2 building.
B - change description	RSCaMa with CLIP ViT-B/32, spatial/temporal state-space fusion and a Transformer language decoder [3]	7 x 7 patch features, 768-dimensional fusion, three fusion layers, eight attention heads and one decoder layer; greedy output capped at 42 tokens.
C - visual evidence	Custom crop and structural witness	Parameter-free OpenCV processing: LAB luminance mean shift, Sobel edges, rectilinear orientation, local anomalies, visibility and registration diagnostics.
D - claim referee	Qwen3-VL-4B-Instruct image-backed judge; Qwen3-4B-Instruct-2507 shared-evidence comparator [4,5]	Approximately 4B language-model scale; original images plus structured evidence for VL, evidence text for the text comparator. Closed accept/reject/defer outputs.

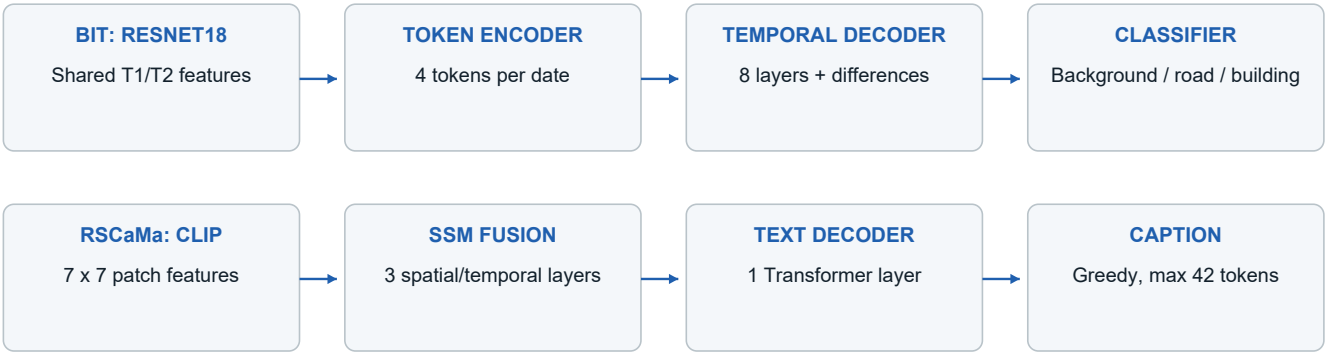


Figure 3. BIT predicts dense class targets. RSCaMa translates paired visual features into text. These modules remain independently replaceable and trainable.

BIT is the detector architecture selection, justified by its compact shared backbone and temporal context mechanism. Detector training uses ImageNet visual initialisation and a new three-class task head. RSCaMa's tested interface consumes two RGB images; detector masks enter the downstream claim-localisation and verification path. This distinction preserves the actual captioner input contract.

The selected captioning comparison includes the released RSCaMa checkpoint and a subset-trained RSCaMa implementation initialised from public CLIP weights. Both referees use the exact checkpoint revisions recorded in the supplied run metadata.

5. Justification and Experimental Evidence

The captioning screen compares a remote-sensing specialist with general vision-language models and deterministic controls. The table uses the **100-pair LEVIR screening population** and automatic mask/reference checks. Strict precision includes all factual claims in its denominator; in-scope precision restricts the evaluated claim population to the supported annotation scope.

Captioning configuration	Strict precision	In-scope precision	BLEU-4
Released RSCaMa	78.7%	88.9%	0.6223
RSCaMa, 20%-budget retrain	79.2%	88.0%	0.5786
Qwen3-VL-4B	58.3%	72.9%	0.0267
SmolVLM2-2.2B	38.6%	48.4%	0.0102
Pixel-difference template	69.0%	69.0%	0.0000
Always no change	40.0%	40.0%	0.2798

The specialist provides the strongest overall captioning baseline in this screen. On Apple M4/MPS at batch one, median caption latency is 0.139 s for RSCaMa and 7.605 s for Qwen3-VL. The two RSCaMa precision intervals overlap; their difference does not establish an improvement from retraining. The released checkpoint uses the original train split and validation-based checkpoint selection. The controlled retrain uses **1,313 train pairs plus 50 development pairs**, separating that comparison from the released caption checkpoint.

These captioning values are the original committed B0 aggregate results. Local implementation checks now reproduce the released RSCaMa values and the two deterministic controls on the same seeded 100-pair screen. The local 20%-budget retrain does **not** reproduce its original aggregate: it scores 76.04% strict precision, 82.02% in-scope precision and 0.5793 BLEU-4, versus 79.17%, 87.96% and 0.5786 in B0. Its local deltas are −3.13 and −5.94 percentage points, and +0.0007 BLEU-4. The screen is a parity check on the same validation population, not new held-out evidence. See the separate implementation evidence report for run provenance, hashes, confidence intervals, vocabulary exposure, timing and the full side-by-side evaluation proof.

The transfer screens use 100 records each from SECOND-CC and WHU-CDC. Released RSCaMa strict precision is **48.0%** and **55.6%**, respectively. This source dependence supports explicit evidence checking and source-specific evaluation. WHU remains a historical transfer screen with its separate usage-rights disposition.

Referee configuration	Completed proxy claims	Raw decision agreement
Qwen3 text, common evidence	150/150	41.3%
Qwen3-VL, common evidence	150/150	45.3%
Qwen3-VL, image-backed	150/150	56.7%

The referee figures use **30 validation pairs**, constructed annotation-proxy claims and raw choices before the 0.8 accept gate. Image-backed Qwen3-VL is the selected visual-referee configuration; the text model supplies a controlled shared-evidence comparison. These measurements are distinct from independently human-rated caption factuality.

BIT complements these choices through dense localisation and a smaller shared backbone. A local two-epoch BIT-R18 pilot now runs on the RTX 4060 Ti: on 100 held-out pairs from the original archive it reached 45.8% binary any-change F1; mean road/building IoU was 26.8%, and 31 of 45 unchanged pairs contained a predicted change pixel. This small pilot verifies the detector path, not BIT's superiority or a converged score. The comparison with the Otsu control, training subset and exact boundaries are in the Sarthak evidence report. Alternative transformer detectors and Siamese CNNs remain in the benchmark registry and have not been run locally.

6. Model Inputs and Outputs

Contract	Representation
Detector input/output	Paired B x 2 x 3 x 256 x 256 tensors; BIT scaling to [-1,1]; B x 3 x 256 x 256 logits and categorical mask
Captioner features	Author CLIP processor: 224-pixel RGB views, CLIP mean/std; temporal patch features and checkpoint-matched word vocabulary
Structured claim	Object, action, optional exact quantity, image-relative location and exact caption source span
Evidence	Original-frame box, padded crops, transform metadata, mask/witness producer IDs, quality flags and fallible observations
Referee/report	Checkpoint chat template, closed decisions, raw response, probability semantics, gated outcome, reason codes and accepted-description text

The RSCaMa retrain builds its vocabulary from training-subset captions and preserves start/end/padding tokens. The released checkpoint uses its matching vocabulary. Qwen uses the publisher tokenizer/processor and temporal-order prompt. Reference captions, human labels and source masks are excluded from inference requests.

7. Training Strategy

Component	Training configuration and checkpoint rule
BIT task training	ImageNet visual initialisation; backbone frozen for five epochs, then trained at 1e-5; temporal/head parameters at 1e-4. AdamW, weight decay 0.01; cosine schedule, 50-epoch cap; batch eight with four-step accumulation.
Detector loss/selection	Equal mixture of class-weighted focal loss (gamma 2; background/road/building weights 0.10/0.45/0.45) and soft Dice. Validation macro-IoU selects checkpoints; patience eight. Paired detector flips preserve masks.
RSCaMa retraining	Public CLIP visual initialisation; temporal fusion and caption decoder trained through the upstream trainability policy. Token cross-entropy, Adam 1e-4, batch 12 pairs, 40-epoch cap, patience eight and seed zero. Development BLEU-4 selects best weights.
Witness and referee	Structural witness uses fixed image-processing rules. Qwen weights remain frozen; prompt, image budget, model revision and the 0.8 operating gate are recorded.

The table specifies the planned task-training configurations; it does not imply that all full schedules completed. The captioner row records the implemented subset-retraining script and its historical hosted result. Sarthak's local BIT run was a two-epoch fit diagnostic over 100 training pairs, with 25 validation pairs and a separate 100-pair test sample; it kept the ImageNet ResNet-18 backbone frozen and is not the 50-epoch schedule above. Its checkpoint and metrics are retained in the local-only run folder described in the Sarthak evidence report.

Colab supplies GPU training/inference, with L4-class evidence from the returned decision sessions. Model workers use separate environments because captioner and referee library versions differ. Sequential residency and stored intermediates control memory without loading the complete model stack simultaneously.

8. Pipeline Implementation and Verification

The pipeline entrypoint accepts two image paths and a model configuration. Separate workers execute BIT, RSCaMa and the referee; the coordinator performs claim extraction, predicted-region matching, crop inspection and final composition. Model checkpoints, upstream source directories and worker interpreters are explicit configuration fields. This keeps the architecture reproducible across Colab, WSL and local environments.

Acceptance requires a valid decision record and the configured operating gate. Missing evidence and invalid probability vectors produce review reasons. Exact counts require independent instance evidence; connected-region counts do not satisfy that requirement. Output descriptions concern visible changes and preserve uncertainty through the defer state.

Verification evidence	Executed scope
Consolidated CPU suite	**83 passing tests** in the current isolated local environment; no skips
Detector	BIT-R18 trained for two epochs on 100 pairs, validated on 25 and evaluated on 100 held-out original-archive pairs; any-change F1 45.8%, mean road/building IoU 26.8%
Detector control	Validation-selected abs-RGB-difference/Otsu control on the same 100 test pairs; any-change F1 16.7%; false alarm on 45/45 unchanged pairs
Visual witness	Nine unit tests; generated fixture rerun 50/50; local two-pair demonstration matched coarse source-mask region presence in 2/3 hand-authored cases
Captioning	Released RSCaMa rerun on 100/100 seeded screening pairs; strict/in-scope precision and BLEU-4 reproduce 78.7%/88.9%/0.6223; other five B0 configurations and both transfer screens remain historical
Decision exports	Historical report records six ZIP exports, five session IDs and 40 distinct image pairs; not re-run locally
Orchestration	CPU test doubles exercise localisation, reference isolation, decision gating, failure review and preservation of existing run directories

The synthetic witness result is a controlled-fixture regression only. Its local average was 3.96 ms per claim; this timing is not comparable with a learned-model latency. In the local real-image diagnostic, the authored road claim was flagged as changed even though its manual crop contained no nonzero source-mask pixels. This is a mismatch, not a verified road detection. Decision-session exports use a frozen Qwen visual witness and remain separate historical evidence.

The detector pilot was the only learned model stage executed locally. Although the coordinator routes the complete A/B/C/D contracts, this run did not produce a learned caption, referee decision or full end-to-end report. RSCaMa and referee scores above are historical screens, not fresh local results. No independent human-rated factuality results are available.

The planned human assessment uses SUPPORTED, CONTRADICTED, UNSUPPORTED and UNVERIFIABLE judgements from two independent raters. Agreement and Cohen's kappa precede third-rater adjudication using the original images and contextual crops. SUPPORTED maps to accept, CONTRADICTED/UNSUPPORTED to reject and UNVERIFIABLE to defer. This measures claim factuality separately from mask overlap, reference-caption similarity and automated proxy decisions. The project has no completed independent human-rating file yet.

The architecture defines the data flow, model choices and planned training controls. The delivered scripts provide model-worker interfaces, the local detector pilot, evidence records and auditable output generation, supported by the consolidated experiment register. The full learned pipeline and the remaining candidate screens are next implementation work.

References

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